

KARL O. MERCADO

Full-Stack Developer

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Remote · available for US / EU working hours

SUMMARY

CS graduate (UPLB) who builds and deploys full-stack apps end-to-end: a RAG document assistant with vector search and streaming AI responses, database-enforced business systems on Postgres row-level security, and a client site scoring 100 on Lighthouse. Looking for a remote role where I can build, deploy, and iterate independently.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, PHP, Java, C#

Frontend: React, Next.js, Tailwind CSS, Vite, HTML5, CSS3

Backend: Node.js, Express, PHP Laravel, .NET 8, REST APIs, Server Actions

Databases: PostgreSQL (RLS, PL/pgSQL, triggers, pgvector), Supabase, MongoDB, Firebase

AI / LLM: Anthropic API, Voyage AI embeddings, RAG pipelines, structured JSON output prompting

DevOps & Tools: Docker, Git, GitHub Actions CI/CD, Vercel, Railway, n8n (self-hosted), Postman

EXPERIENCE

Web Developer (Freelance) — Peaks & Pedals Transient Jan – Feb 2026

- Designed and deployed a production vacation rental website scoring Lighthouse 100 desktop / 95 mobile. Next.js 15, React 19, Tailwind CSS 4; SEO via JSON-LD, OpenGraph, and AVIF/WebP image optimization.
- Owned the full project lifecycle: client requirements, design iteration, domain setup, and Vercel deployment. Live: peaksnpedals.com

Software Engineer Intern — Argon Software Jun – Aug 2024

- Engineered a voucher management module in PHP Laravel (creation, assignment, validation) with role-based access; refactored Blade templates and CSS for responsive landing pages.

Web Developer (Contract) — Municipal Government Office May – Jun 2024

- Built and deployed a responsive WordPress government website organizing 50+ pages of public service content.

PROJECTS

Site Manager for Construction Company Next.js 16 · React 19 · TypeScript · Supabase · Postgres RLS

- Built and deployed a material-tracking web app for a construction firm, replacing Excel and paper logbooks. Append-only stock ledger where all balances are computed Postgres views; row-level security hides prices and budgets from field roles at the database layer, verified by automated per-role API tests. Live demo available on request (repo private).

Municipal Citation Monitoring Desk React · TypeScript · Supabase · PL/pgSQL

- Citation-tracking system encoding a municipal ordinance's rules as Postgres triggers: a trigger-enforced ticket state machine, holiday-aware due-date computation, per-role RLS on every table, full audit log, and an offline draft queue for field enforcers. Walkthrough available on request.

MayHapotTabi — AI Document Chat React · TypeScript · Node.js · Supabase pgvector · Anthropic API

- Full RAG pipeline: PDF ingestion > 400-token chunking > Voyage AI embeddings > pgvector cosine search > Claude streaming responses. Per-user isolation via Supabase RLS; Docker multi-stage builds and GitHub Actions CI. [Code](#) · [Live](#)

UPCAT Reviewer (CEER) Next.js 16 · React 19 · TypeScript · Supabase

- Shipped a live mobile-first quiz platform. Server-side grading; the answer key never reaches the client (no SELECT on questions, reads go through SECURITY DEFINER RPCs), with RLS and column grants enforcing per-user privacy. [Code](#) · [Live](#)

n8n Automation Workflows n8n · Claude API · HubSpot · Gmail · Slack · Google Sheets

- Four public importable workflows: AI lead qualification with idempotent HubSpot upserts, KB-grounded support triage with fail-safe escalation, and a content pipeline with Slack-button approval over a 48h wait/resume webhook. [Code](#)

EDUCATION

BS Computer Science — University of the Philippines Los Baños 2019 – 2025